

Ideal Versus Non-Ideal Models of Active Filters

Miles Feheley
 Dept. of Electrical Engineering
 University of Vermont
 Burlington, United States of America
 milesfeheley@gmail.com

Abstract—This report aims to explore the differences that occur when modeling operational amplifiers, or op-amps, as ideal or non-ideal. Modeling op-amps as ideal allows for easier calculation, however it ignores many imperfections that can cause a design to fail if not addressed. To visualize the different behaviors, an active inverting high-pass filter was modeled with three different non-ideal op-amps as well as an ideal op-amp. The three non-ideal op-amp models utilized were the LM741, the TL072, and the OPA2134. These models were then run through a numerical analysis, where their response to a range of frequencies was examined. Results from this analysis show that non-ideal op-amps behave similarly within a certain bandwidth of frequencies, however, this behavior varies drastically outside of that range. This drastic variation in behavior is an exemplar of the risk that can occur in modeling op-amps as ideal in circuits operating at high frequencies.

I. INTRODUCTION

Operational amplifiers are used in an extremely wide variety of applications in modern technology. This report aims to explore the benefits and drawbacks of two different methods of modeling an op-amp for numerical analysis. By increasing the accuracy of numerical models, we are more effectively able to predict the behavior of systems before testing them. Further analysis of this subject can also lead to improvements in op-amp designs in the future, allowing for further development in op-amp usage.

II. METHOD OF ANALYSIS

A. Circuit Design and Topology

To explore the variation in behavior in a practical application, the op-amps are modeled within an active inverting high-pass filter. The circuit topology utilized is shown below in Figure 1.

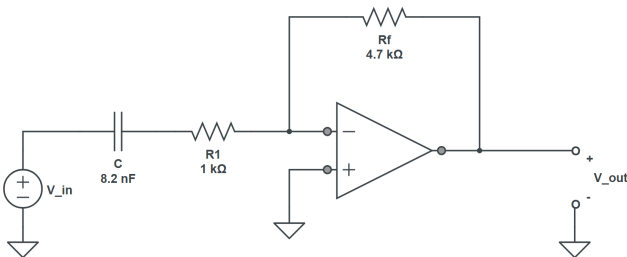


Fig. 1. Base Circuit Topology

Component values utilized in this design are listed below in Table I as well as their symbol and name in reference to Figure 1.

TABLE I
COMPONENT NAMES AND VALUES

Name	Symbol	Value	Units
Input Capacitor	C	8.2	nF
Input Resistor	R_1	1	kΩ
Feedback Resistor	R_f	4.7	kΩ

This filter was designed to attenuate frequencies below 20 kHz. These component values were picked specifically to accomplish this goal. The high-pass filter cutoff frequency was calculated using Equation 1 shown below.

$$\omega_c = \frac{1}{R_1 C} \quad (1)$$

This op-amp application is used in many systems, most with a notable application being audio systems. Many modern equalizers utilize active high-pass filters similar to this to filter audio for large scale sound systems.

B. Assumptions In An Ideal Model

The ideal op-amp model assumes certain properties that are untrue in a practical application. A rough model of realistic op-amps can be seen below in Figure 2.

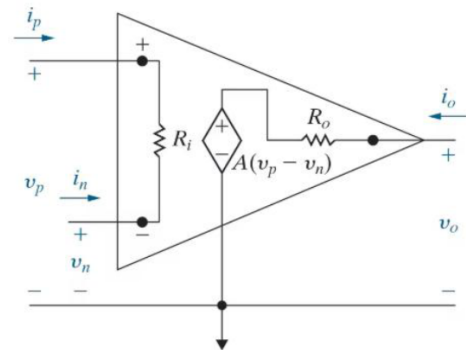


Fig. 2. Op-Amp Realistic Model

In the ideal model, we assume infinite gain, infinite input impedance, zero input current, zero output impedance, and

zero input offset voltage. We are able to assume that the inverting and non-inverting inputs are at an equal voltage, and that the signal encounters no impedance within the op-amp. We also assume the op-amp can amplify signals to any desired levels. This allows for simplifications in circuit solving, however, it also introduces inaccuracies to analytical models.

In the realistic model, the op-amp will be replaced with a set of resistors and a dependent voltage source. While this is still a simplification, this model allows for much more accurate results in behavioral analysis. An image of the realistic model circuit topology is shown below in Figure 3.

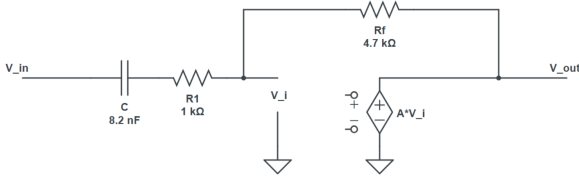


Fig. 3. Circuit Topology for Realistic Models

C. Analytical Software

To model these filters, Python was used. Doing this allowed for iteration through a set of predefined component and op-amp properties. Sample code utilized can be seen below in Code 1. The full script utilized to model the op-amps can be seen in the appendices.

```
1 num = [-Rf*C, 0]
2 den = [R1*C, 1]
3 system = signal.TransferFunction(num, den)
4 _, mag, phase = signal.bode(system, w)
5
6 ax1.semilogx(frequencies, mag)
7 ax2.semilogx(frequencies, phase)
```

Code 1. Example of Python code.

III. ANALYTICAL SOLUTIONS

To obtain an analytical model of the differing models, the circuits were analyzed in the Laplace space. This allowed for simplified calculations within time-variant systems. For each model, the transfer function, defined below in Equation 2, was determined.

$$T(s) = \frac{V_{in}}{V_{out}} \quad (2)$$

A. Ideal Model

Due to assumptions made for the op-amp, the transfer function derivation for the ideal model is fairly simplistic. The ideal model transfer function is shown below in Equation 3.

$$T(s) = -\frac{R_2}{\frac{1}{sC} + R_1} = -\frac{sR_2C}{1 + sR_1C} \quad (3)$$

B. Non-Ideal Model

The non-ideal model required a more complicated derivation as the ideal assumptions could no longer be made. To determine op-amp properties, research was conducted into each realistic model to determine electrical properties pertinent to frequency response. These properties were recorded and utilized in numerical analysis. While each non-ideal model had differing electrical properties, the symbolic representation of the model remained consistent. The two properties of interest in each op amp were A_0 , the DC gain, and ω_c , the cutoff frequency of the op-amp. Using these values, we can define the transfer function of the non ideal op-amp as seen in Equation 4.

$$A(s) = \frac{A_0}{\frac{1}{\omega_c}s + 1} \quad (4)$$

This however is just the transfer function for the op-amp itself. The transfer function of the entire non-ideal model is shown below in Equation 5

$$T(s) = \frac{-sA_0CR_f\omega_c}{s^2C(R_f + R_1) + s(CR_1A_0\omega_c + CR_1\omega_c + 1) + CR_f\omega_c + A_0\omega_c + \omega_c} \quad (5)$$

IV. NUMERICAL RESULTS

Using the transfer functions defined for each model in the analytical solutions, we are able to plot the frequency responses of each op-amp. This numerical analysis will provide us with a good indication of each model's magnitude and phase response to a range of frequencies.

For each op-amp used in this analysis, we are able to obtain an open-loop large-signal differential voltage amplification vs. frequency graph. From this, we are able to determine the values of A_0 and ω_c for each model. These plots can be seen below in Figures 4, 5, and 6.

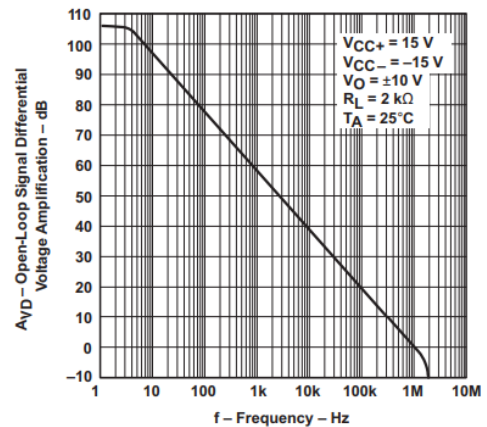


Fig. 4. LM741 Open-Loop Large-Signal Differential Voltage Amplification vs. Frequency

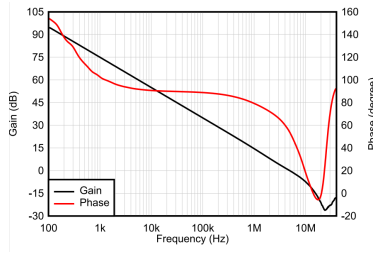


Fig. 5. TL072 Open-Loop Large-Signal Differential Voltage Amplification vs. Frequency

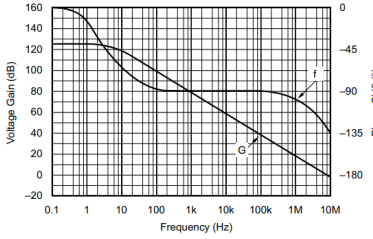


Fig. 6. OPA2134 Open-Loop Large-Signal Differential Voltage Amplification vs. Frequency

From these graphs, we are able to determine the required values, which can be seen below in Table II.

TABLE II
OP-AMP DC GAIN AND CUTOFF FREQUENCY VALUES

Name	DC Gain	Cutoff Frequency (Hz)
LM741	$2 \cdot 10^5$	25
TL072	$1 \cdot 10^6$	200
OPA2134	$1 \cdot 10^6$	10

Plugging these values into the transfer function defined in Equation 5, as well as the values of the ideal model in Equation 3, we are able to generate the Bode diagram shown below in Figure 7.

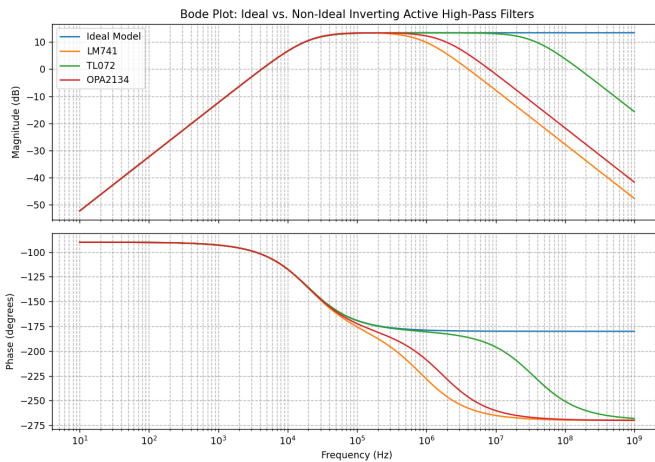


Fig. 7. Frequency Response of Ideal and Non-Ideal Models

V. RESULTS AND DISCUSSION

The responses found in this study align well with expected trends and values. We are able to see the band-pass behavior that was expected from the non-ideal model, and the continuous high-frequency pass that was expected from the ideal model. The desired cut-off frequencies can also be seen in the results. Around 20 kHz, each model reaches its lower cutoff frequency. This is expected since the high-pass filter elements are the same for each model. Once all models reach the maximum value, the difference between each op-amp starts to show. As a result of the assumptions made for the ideal op-amp, that model is able to pass any frequency above the high-pass cutoff. The non-ideal op amps begin to attenuate very high frequencies on the basis of their DC gain and cutoff frequency values. By comparing Table II and Figure 7, we can see that an increase in DC gain and cutoff frequency increases the bandwidth of the filter. The TL072 has the greatest cutoff frequency and DC gain, so it begins attenuating later. The OPA2134 has a lower cutoff frequency than the LM741, however, its DC gain is larger by a greater factor, so it has a larger bandwidth.

The phase diagram also provides a good reference for the applicability of this filter design to physical systems, particularly audio signal processing. The ideal op-amp provides a perfect audio filter. Any audio signals below the cutoff frequency are at a 90-degree phase angle and will not be audible. As the system transitions to higher frequencies, the signal will become audible until it reaches its maximum value at a 180-degree phase angle. This audio will produce no problems for listeners unless a non-inverted audio signal is combined. As frequencies increase in the non-ideal models, the phase angle begins to diverge from the desired 180-degree angle. As it approaches 270 degrees, the audio signal will become attenuated again. The order in which the op-amps start to diverge from the 180-degree angle is the same as the order in which they diverge from their maximum magnitude.

If a band-pass filter topology cannot be implemented into a design, the cutoff of an op-amp can be used to your advantage. By carefully selecting the type of op-amp used in your design, you can ensure signals above a certain frequency become attenuated. For applications where a larger bandwidth is required, the TL072 is the best choice of those analyzed. For applications looking for a shorter bandwidth, the LM741 or OPA2134 is a better choice.

VI. PERSONAL APPLICATION

This active filter model can be extremely helpful in an audio formatting setting. In the past, I have done a lot of work with setting up and tuning sound equipment. As I have come to understand the function and design concepts behind passive and active filters, I have a better appreciation for not only the equipment I used, but how projects I undertook could have been further improved. This model is an excellent example of an unorthodox method of handling unwanted audio signals in a live setting.

The inverting operation of this op-amp can be used to your advantage when things are going wrong with audio. Because it causes a 180-degree phase shift, when the inverted signal gets combined back with the unfiltered audio signal, the frequencies within the band-pass range become negated. An illustration of this concept is shown below in Figure 8.

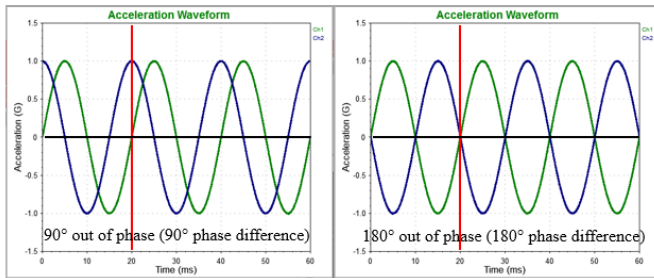


Fig. 8. 90-Degree and 180-Degree Signal Offsets

When doing live audio mixing, oftentimes, unwanted frequencies can arise. Implementing this filter in the described manner would enable you to negate the unwanted frequencies without adjusting settings or reformatting filters.

This concept isn't exactly new, however. This is the same concept used in noise-cancellation technology, such as headphones, cars, planes, and industrial equipment. Noise-canceling systems begin with an input microphone that records background noise. This signal is then passed through an inverting amplifier, and an opposing sound wave is produced to cancel out noise. This concept is often referred to as Active Noise Control (ANC) or Active Noise Reduction (ANR).

To further explore this idea, the lower cutoff frequency of the filter could be made adjustable by implementing variable components such as a potentiometer. This would allow you to adjust the resistance of your input resistor, which would directly affect your cutoff frequency as shown in Equation 1. Applying ANR concepts to a variable filter would allow for selective frequency reduction, as opposed to the general noise reduction utilized in most ANR systems.

If this concept were implemented into an audio system, the analytical and numerical modeling conducted in this report would be fairly accurate to real values. Speakers attached to the output would not produce a resistive load, but instead a reactive load. This would require further care to preserve a 180-degree offset in selected frequencies, but modeling from this report would remain relatively accurate.

REFERENCES

- [1] Texas Instruments Incorporated. (2018). MA741 General-Purpose Operational Amplifiers Technical Documents. In Texas Instruments (Vols. SLOS094G–SLOS094G). <https://www.ti.com/lit/ds/symlink/ua741.pdf>
- [2] Texas Instruments Incorporated. (2023). TL07XX Low-Noise FET-Input Operational Amplifiers [Report]. <https://www.ti.com/lit/ds/symlink/tl074.pdf?ts=1745990864896refurl=https%253A%252F%252Fwww.ti.com%252FTL074>
- [3] Texas Instruments Incorporated. (2024). OPAX134 High-Performance, SoundPlus™ Audio Operational Amplifiers. In OPA134, OPA2134, OPA4134 [Technical report]. <https://www.ti.com/lit/ds/symlink/opa2134.pdf?ts=1746367188739refurl=https%253A%252F%252Fwww.mouser.ca%252F>
- [4] VRU. (2023, November 11). Phase of a Sine Wave - Vibration Testing Basics - VRU. <https://vru.vibrationresearch.com/lesson/phase-and-sine/>

VII. APPENDICES

A. Full Code

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 import scipy.signal as signal
4
5 C_list = [8.2e-9, 8.2e-9, 8.2e-9, 8.2e-9] # Capacitance in Farads
6 Rl_list = [1e3, 1e3, 1e3, 1e3] # Input resistors in Ohms
7 Rf_list = [4.7e3, 4.7e3, 4.7e3, 4.7e3] # Feedback resistors in Ohms
8
9 A0_list = [0, 2e5, 1e6, 1e6] # DC gains (unitless)
10 fc_list = [0, 25, 200, 10] # Cutoff frequencies in Hz
11 wc_list = [2 * np.pi * fc for fc in fc_list] # Convert to rad/s
12
13 frequencies = np.logspace(1, 9, 1000) # 10 Hz to 1 GHz
14 w = 2 * np.pi * frequencies # rad/s
15
16 fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 7), sharex=True)
17 for i in range(4):
18     C = C_list[i]
19     Rl = Rl_list[i]
20     Rf = Rf_list[i] # Define values for current iteration
21     A0 = A0_list[i]
22     wc = wc_list[i]
23     if i == 0:
24         num = [-Rf*C, 0]
25         den = [Rl*C, 1]
26
27         system = signal.TransferFunction(num, den)
28         _, mag, phase = signal.bode(system, w)
29     else:
30         num = [-A0 * wc * C * Rf, 0]
31         den = [C*(Rf + Rl), (C*Rl*A0*wc + 1), (C*Rf*wc + A0*wc + wc)]
32
33         system = signal.TransferFunction(num, den)
34         _, mag, phase = signal.bode(system, w)
35     if i == 0:
36         ax1.semilogx(frequencies, mag, label=f'Ideal Model')
37         ax2.semilogx(frequencies, phase)
38     elif i == 1:
39         ax1.semilogx(frequencies, mag, label=f'LM741')
40         ax2.semilogx(frequencies, phase)
41     elif i == 2:
42         ax1.semilogx(frequencies, mag, label=f'TL072')
43         ax2.semilogx(frequencies, phase)
44     elif i == 3:
45         ax1.semilogx(frequencies, mag, label=f'OPA2134')
46         ax2.semilogx(frequencies, phase)
47
48 ax1.set_ylabel('Magnitude (dB)')
49 ax1.set_title('Bode Plot: Ideal vs. Non-Ideal Inverting Active High-Pass Filters')
50 ax1.legend()
51 ax1.grid(True, which='both', linestyle='--')
52
53 ax2.set_ylabel('Phase (degrees)')
54 ax2.set_xlabel('Frequency (Hz)')
55 ax2.grid(True, which='both', linestyle='--')
56
57 plt.tight_layout()
58 plt.show()
```

Code 2. Example of Python code.